

Jialian Wu

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Current Employment

AMD GenAI

San Jose, CA

Senior Research Scientist (Senior MTS)

June 2025 – Present

Research Scientist (MTS)

Apr 2024 – June 2025

Focus: Research and train open LLMs and Coding Agents.

Core contributor for releases:

1) **Instella** [\[Tech Report\]](#): a 3B language model pre-trained and post-trained with 4T tokens.

2) **Instella-Long (Lead)** [\[Blog\]](#): a long-context language model trained with long-context continued pre-training and post-training, surpassing Qwen, Phi, and Gemma.

3) **AMD-OLMo** [\[Blog\]](#): AMD's first billion-parameter language model trained from scratch with 1T tokens.

Education

State University of New York at Buffalo, USA

Aug 2019 - June 2023

Ph.D. in Computer Science and Engineering

Advisor: Dr. Junsong Yuan

Awards: Best Graduate Research Award, Best First Year Achiever Award

Tianjin University, China

Bachelor in Electronic Engineering, GPA: 3.85/4.0

Sept 2014 – July 2018

Publications (Full List at [Google Scholar](#)) ([†]student mentorship)

- Kaijie Zhu, Yuzhou Nie, Yijiang Li, Yiming Huang, **Jialian Wu**[†], Wenbo Guo, et al. “TermiGen: High-Fidelity Environment and Robust Trajectory Synthesis for Terminal Agents”, arXiv 2026.
- Jingyang Lin, **Jialian Wu**[†], Jiang Liu, Ximeng Sun, Ze Wang, Xiaodong Yu, Jiebo Luo, Zicheng Liu, Emad Barsoum, “VideoSeek: Long-Horizon Video Agent with Tool-Guided Seeking”, **CVPR**, 2026.
- Jialian Wu**, Jiang Liu, Sudhanshu Ranjan, Xiaodong Yu, Gowtham Ramesh, Prakamya Mishra, Zicheng Liu, et al. “Instella-Long: A Fully Open Language Model with Long-Context Capability”, [AMD Blog](#), 2025.
- Jiang Liu, **Jialian Wu**, Xiaodong Yu, Prakamya Mishra, Sudhanshu Ranjan, Zicheng Liu, et al. “Instella: New State-of-the-art Fully Open 3B Language Models”, [Tech Report](#), 2025.
- Zhikun Xu, Xiaodong Yu, Ben Zhou, Jiang Liu, **Jialian Wu**, Ze Wang, Ximeng Sun, Hao Chen, Zicheng Liu, “Reliable Use of Lemmas via Eligibility Reasoning and Section-Aware Reinforcement Learning”, **ACL**, 2026.
- Chao Huang, Zeliang Zhang, Jiang Liu, Ximeng Sun, **Jialian Wu**, Xiaodong Yu, Ze Wang, Chenliang Xu, Emad Barsoum, Zicheng Liu, “DRIFT: Transferring Reasoning Priors for Efficient MLLM Fine-Tuning”, **ACL Findings**, 2026.
- Han Wang, Xiaodong Yu, **Jialian Wu**, Jiang Liu, Ximeng Sun, Mohit Bansal, Zicheng Liu, “Stabilizing Efficient Reasoning with Step-Level Advantage Selection”, **ACL Findings**, 2026.
- Bangzheng Li, Ximeng Sun, Jiang Liu, Ze Wang, **Jialian Wu**, Xiaodong Yu, Hao Chen, Emad Barsoum, Muhao Chen, Zicheng Liu “Latent Visual Reasoning”, **ICLR**, 2026.

9. Xingrui Wang, Jiang Liu, Chao Huang, Xiaodong Yu, Ze Wang, Ximeng Sun, **Jialian Wu**, Alan Yuille, Emad Barsoum, Zicheng Liu “XModBench: Benchmarking Cross-Modal Capabilities and Consistency in Omni-Language Models”, **ICLR**, 2026.
10. Yuxiang Guo, Jiang Liu, Ze Wang, Hao Chen, Ximeng Sun, Yang Zhao, **Jialian Wu**, Xiaodong Yu, Zicheng Liu, Emad Barsoum “ImageDoctor: Diagnosing Text-to-Image Generation via Grounded Image Reasoning”, **ICLR**, 2026.
11. Yujian Liu, Ze Wang, Hao Chen, Ximeng Sun, Xiaodong Yu, **Jialian Wu**, Jiang Liu, Emad Barsoum, Zicheng Liu, Shiyu Chang “Learning from Online Videos at Inference Time for Computer-Use Agents”, **TMLR**, 2026.
12. Jingyang Lin, **Jialian Wu**[†], Ximeng Sun, Ze Wang, Jiang Liu, Yusheng Su, Xiaodong Yu, Hao Chen, Jiebo Luo, Zicheng Liu, Emad Barsoum, “Unleashing Hour-Scale Video Training for Long Video-Language Understanding”, **NeurIPS (Spotlight)**, 2025.
13. Yuzhen Zhou, Jiajun Li, Yusheng Su, Gowtham Ramesh, Zilin Zhu, Xiang Long, Chenyang Zhao, Jin Pan, Xiaodong Yu, Ze Wang, Kangrui Du, **Jialian Wu**, Ximeng Sun, Jiang Liu, Qiaolin Yu, Hao Chen, Zicheng Liu, Emad Barsoum, “APRIL: Active Partial Rollouts in Reinforcement Learning to Tame Long-tail Generation”, *arXiv* 2025.
14. Yufan Zhuang, Xiaodong Yu, **Jialian Wu**, Ximeng Sun, Ze Wang, Jiang Liu, Yusheng Su, Jingbo Shang, Zicheng Liu, Emad Barsoum, “Self-Taught Agentic Long Context Understanding”, **ACL**, 2025.
15. Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, **Jialian Wu**, Xiaodong Yu, Jiang Liu, Zicheng Liu, Emad Barsoum, “Agent Laboratory: Using LLM Agents as Research Assistants”, **EMNLP Findings**, 2025.
16. Prakamya Mishra, Jiang Liu, **Jialian Wu**, Xiaodong Yu, Zicheng Liu, Emad Barsoum, “TTT-Bench: A Benchmark for Evaluating Reasoning Ability with Simple and Novel Tic-Tac-Toe-style Games”, **EMNLP**, 2025.
17. Jiang Liu, **Jialian Wu**, Prakamya Mishra, Zicheng Liu, et al. “Introducing the First AMD 1B Language Models: AMD OLMo”, **AMD Blog**, 2024.
18. **Jialian Wu**, Jianfeng Wang, Zhengyuan Yang, Zhe Gan, Zicheng Liu, Junsong Yuan, and Lijuan Wang, “GRiT: A Generative Region-to-text Transformer for Object Understanding”, **ECCV**, 2024.
19. **Jialian Wu**, Sudhir Yarram, Hui Liang, Tian Lan, Junsong Yuan, Jayan Eledath, and Gerard Medioni, “Efficient Video Instance Segmentation via Tracklet Query and Proposal”, **CVPR**, 2022.
20. **Jialian Wu**, Jiale Cao, Liangchen Song, Yu Wang, Ming Yang, and Junsong Yuan, “Track to Detect and Segment: An Online Multi-Object Tracker”, **CVPR**, 2021.
21. **Jialian Wu**, Liangchen Song, Qian Zhang, Ming Yang, and Junsong Yuan, “ForestDet: Large-Vocabulary Long-Tailed Object Detection and Instance Segmentation”, **TMM**, 2021.
22. Liangchen Song, **Jialian Wu**, Ming Yang, Qian Zhang, Yuan Li, and Junsong Yuan, “Stacked Homography Transformations for Multi-View Pedestrian Detection”, **ICCV (Oral)**, 2021.
23. Liangchen Song, **Jialian Wu**, Ming Yang, Qian Zhang, Yuan Li, and Junsong Yuan, “Handling Difficult Labels for Multi-label Image Classification via Uncertainty Distillation”, **ACM MM**, 2021.
24. Liangchen Song, **Jialian Wu**, Ming Yang, Qian Zhang, Yuan Li, and Junsong Yuan, “Robust Knowledge Transfer via Hybrid Forward on the Teacher-Student Model”, **AAAI**, 2021.
25. **Jialian Wu**, Chunluan Zhou, Ming Yang, Qian Zhang, Yuan Li, and Junsong Yuan, “Temporal-Context Enhanced Detection of Heavily Occluded Pedestrians”, **CVPR**, 2020.
26. **Jialian Wu**, Liangchen Song, Tiancai Wang, Qian Zhang, and Junsong Yuan, “Forest R-CNN: Large-Vocabulary Long-Tailed Object Detection and Instance Segmentation”, **ACM MM**, 2020.

27. **Jialian Wu**, Chunluan Zhou, Qian Zhang, Ming Yang, and Junsong Yuan, “Self-Mimic Learning for Small-scale Pedestrian Detection”, **ACM MM**, 2020.

Previous Experience

Research Scientist, Qualcomm

Multimodal Understanding in Videos

June 2023 - April 2024

San Diego, CA, USA

Applied Scientist Intern, Amazon

Amazon Go Team, Mentors: Dr. Tian Lan, Dr. Hui Liang

Aug 2022 - Oct 2022

Seattle, WA, USA

- *Project: Weakly Supervised Video Instance Segmentation*

Research Intern, Microsoft

Microsoft Azure AI Team

May 2022 - Aug 2022

Redmond, WA, USA

Mentors: Dr. Jianfeng Wang, Dr. Zhe Gan, Dr. Lijuan Wang, Dr. Zhengyuan Yang, Dr. Zicheng Liu

Project: Multimodal Understanding at Region-level

Applied Scientist Intern, Amazon

Amazon Go Team, Mentors: Dr. Tian Lan, Dr. Hui Liang

May 2021 - Aug 2021

Seattle, WA, USA

- *Project: Video Instance Segmentation*

Research Intern, Horizon Robotics

Autonomous Driving Perception Team, Mentor: Dr. Yu Wang Dr. Ming Yang

May 2020 - Aug 2020

Cupertino, CA, USA

- *Project: Multi-Object Tracking*

Research Intern, Horizon Robotics

Mentor: Dr. Qian Zhang, Dr. Ming Yang

May 2018 - July 2019

Beijing, China

- *Project: Occluded and Small Pedestrian Detection*

AWARDS & HONORS

1. Best Graduate Research Award, State University of New York at Buffalo, 2022.
2. Best First Year Achiever Award, State University of New York at Buffalo, 2020.
3. Outstanding Bachelor Thesis, Tianjin University, 2018.
4. Tianjin City Fellowship, 2016.

Professional Services

Conference Reviewer: CVPR 2020/2021/2022/2024/2025/2026, ICLR 2024/2025, ICML 2024, NeurIPS 2023/2025, ICCV 2021/2023, ECCV 2022/2024, ACL Rolling Review 2026, AAAI 2021/2022/2023, IJCAI 2021/2022, WACV 2021/2022, ICASSP 2021, etc

Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Image Processing, IEEE Transactions on Multimedia, IEEE Transactions on Circuits and Systems for Video Technology, Neurocomputing, Machine Vision and Applications, Neurocomputing, The Visual Computer

Teaching Assistant:

- CSE573: Computer Vision and Image Processing, Fall 2019.
- CSE191: Discrete Structures, Spring 2020.